System 126 (2024) 103478

Contents lists available at ScienceDirect

System

journal homepage: www.elsevier.com/locate/system

A systematic review of research on emotional artificial intelligence in English language education

Yuhan Liu [a], Heng Zhang [a], Meilin Jiang [a], Juanjuan Chen [b], Minhong Wang [a,*]

[a] Faculty of Education, The University of Hong Kong, Hong Kong
[b] College of Education, Zhejiang University, China

ARTICLE INFO

Keywords:
Language education
English as a foreign language (EFL)
Artificial intelligence
Emotion
Systematic review

ABSTRACT

In learning English as a foreign language (EFL), students often experience foreign language anxiety. Artificial intelligence (AI) applications that provide emotional support and/or create emotional impacts on student learning, so-called emotional AI applications, have received increased attention. However, there is a lack of a systematic review of studies on emotional AI in EFL education. This paper presents a systematic review of research in this field. The results reveal five affordances of emotional AI in EFL education, namely (1) enabling human-like conversations, (2) providing personalized real-time feedback or instructions, (3) translating images into English text, (4) generating personalized learning content and tasks, and (5) recognizing and analyzing emotions. The first three affordances are more frequently used and have shown promising effects on improving students' behavioral, cognitive, and affective learning outcomes. Moreover, the findings reveal that emotional support is often integrated with cognitive support; providing emotional support alone may not be enough to support student learning. Meanwhile, providing cognitive support alone can enhance both affective and cognitive learning outcomes. Finally, attention should be paid to the factors that might influence the adoption and effects of emotional AI in EFL education.

1. Introduction

In educational contexts, students often experience cognitive challenges and negative emotions. It is thus important to provide both cognitive and emotional support for students. Artificial intelligence (AI) applications that provide emotional support and/or create emotional impacts on student learning, so-called *emotional AI applications*, have received increased attention. When learning English as a foreign language (EFL), students often experience foreign language anxiety, which may impede their academic achievement (Chen & Lee, 2011; Pekrun et al., 2017). Accordingly, using emotional AI applications to enhance language teaching and learning has been increasingly explored in EFL education (Chen & Lee, 2011; Huang et al., 2023). Currently, the research on emotional AI in EFL education is in a state of fragmentation, hindering a comprehensive understanding of the field. This paper presents a systematic review of published studies on emotional AI in EFL education, aiming to synthesize existing findings and provide suggestions for future research and practice in this field.

* Corresponding author. Faculty of Education, The University of Hong Kong, Pokfulam Road, Hong Kong.
E-mail addresses: yhliu19@connect.hku.hk (Y. Liu), zhangheg@connect.hku.hk (H. Zhang), mljiang@connect.hku.hk (M. Jiang), juanjuanchen@zju.edu.cn (J. Chen), magwang@hku.hk (M. Wang).

https://doi.org/10.1016/j.system.2024.103478
Received 31 January 2024; Received in revised form 2 September 2024; Accepted 5 September 2024
Available online 9 September 2024
0346-251X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1.1. AI in education

AI refers to the ability of computers or computer-based systems to perform tasks in a way that mimics human intelligence, such as acquiring and processing information and knowledge, reasoning, and problem solving (Russell & Norvig, 2016). AI may also refer to the ability of computers to extend human intelligence, for example by enabling rapid processing of a large amount of data in various forms to recognize patterns and make predictions from the data through machine learning and data mining techniques (Chen et al., 2022).

In recent years, AI has received significant attention in the field of education (Chen et al., 2020; Zawacki-Richter et al., 2019). Intelligent tutoring systems have been widely explored and applied to support student learning (VanLehn, 2011). Such kind of systems often involve pedagogical agents that can provide immediate and personalized feedback to learners through modeling student profiles, analyzing individual performance, and imitating human tutors' instructional strategies. Besides, some applications can generate personalized learning content and tasks to support personalized adaptive learning (Rus et al., 2013; Sikström et al., 2022).

AI in education is closely associated with educational data mining and learning analytics, which focus on the analysis of various educational data by using statistical methods as well as data mining techniques (Chen et al., 2020). Based on automated analysis of student performance data (e.g., class attendance, assignment submission), such kind of AI applications can help identify potential problems of students, predict student performance, and generate relevant alerts or advice to support student learning and enhance administrative efficiency in education. More recently, researchers have discussed the use of AI-based conversational agents that can imitate natural language and facilitate human-like conversations to engage students in meaningful dialogue and interactive learning (Rus et al., 2013).

1.2. AI in language education

With the promotion of AI in education, AI has shown a transformative force in language education (Huang et al., 2023; Sharadgah & Sa'di, 2022; Smutny & Schreiberova, 2020). AI-based applications have been explored to support language education in multiple aspects such as writing, speaking, reading, vocabulary, grammar, and translation. They provide conducive and less intimidating environments for students to practice language skills through repetitive exercises. Moreover, AI-assisted assessment and feedback systems have been explored to evaluate students' writing and generate personalized feedback in multiple aspects (including wording, content, grammar, syntax, and structure); they help students correct errors and improve writing performance (Strobl et al., 2019; Yang & Kyun, 2022). Other AI-based tools are developed to support students' reading and comprehension. For example, Hsiao and Chang (2023) developed a reading tool that can generate a list of words with simplified explanations based on the reading materials and the student's proficiency level; it can also offer explanations of ambiguous words or present context-sensitive meanings of words.

More recently, conversational agents or chatbots have been increasingly applied in EFL education (Huang et al., 2023; Smutny & Schreiberova, 2020; Yang & Kyun, 2022). They can offer a variety of language learning activities such as conversational practices and vocabulary and grammar exercises and provide instant responses to student queries. In particular, they can use voice-based communication to support speaking or pronunciation practice with helpful feedback. By providing autonomous, affective, and coherent communications with students, conversational agents or chatbots can enhance students' performance as well as increase their interest and engagement, reduce foreign language and speech-related anxiety, and encourage students' social presence (Huang et al., 2022; Yang & Kyun, 2022).

1.3. Emotional AI in EFL education

Emotions are prevalent in teaching and learning processes. Research has revealed the profound impact of emotions on learning performance such as attention, understanding, memory, and task performance (Brand et al., 2007; Feidakis, 2016). In particular, anxiety is identified as a distinctive emotion due to its significant impact on learning performance. In the field of second or foreign language education, there is considerable evidence of positive relationships between positive emotions and academic success (Oxford, 2015; Plonsky et al., 2022). People who have poor language learning abilities tend to experience foreign language anxiety, which has shown an adverse effect on learning performance (Chen & Lee, 2011; Hu et al., 2024).

With new advances in technology, AI applications that provide emotional support and/or create emotional impacts on student learning, so-called *emotional AI applications*, have received increased attention. Research on emotional AI in EFL education has focused on (a) utilizing AI applications that can provide emotional support for learners, and (b) investigating the emotional impacts of AI applications by assessing affective learning outcomes. First, researchers have explored the use of AI applications that can provide emotional support to EFL students, for example by generating facial expressions, gestures, emotion-related recommendations, and empathetic instructions to reduce anxiety (Chen & Lee, 2011; Feidakis, 2016; Xu, 2018). Student emotions can be recognized through the analysis of facial expressions, tone of voice, physiological signs (e.g., human pulse, blood pressure) and observable behavior (Chen & Lee, 2011). Given that many EFL learners have foreign language anxiety, emotional AI applications like affective tutoring systems have been applied to EFL education (Huang et al., 2023). They can help identify students' emotional needs to allow for more personalized and empathetic instructions. Moreover, conversational agents can make gestures and facial expressions to reduce student anxiety during the study (Ericsson et al., 2023; Ericsson & Johansson, 2023; Lee et al., 2011). The increasing use of conversational agents in language education can also inspire empathetic engagement and allow students to actively engage in learning without fear of being judged (Ayedoun et al., 2019; Lee et al., 2011; Rahman & Tomy, 2023).

Second, researchers have investigated the emotional impacts of AI applications on EFL education by assessing students' affective

learning outcomes such as enjoyment, anxiety, self-efficacy, and confidence (Ericsson et al., 2023; Hsu et al., 2023; Nazari et al., 2021; Rahman & Tomy, 2023; Sargazi Moghadam et al., 2023; Shin & Lee, 2021). Although many AI applications focus on supporting students' cognitive development (e.g., writing, speaking, reading, vocabulary), they have shown promising effects on improving students' emotional experience. For example, Nazari et al. (2021) found that using an AI digital writing tool to provide formative feedback and assessment on student performance can enhance students' enjoyment and self-efficacy and decrease their anxiety, in addition to improving writing performance. Such findings can be explained by the reciprocal relationships between cognition and emotion (Phelps, 2006). When performing complex tasks, students often experience negative emotions such as anxiety, which may affect their cognitive process and learning achievements (Pekrun et al., 2017). In such a context, providing students with cognitive support to solve the task may improve their thinking process as well as reduce their anxiety and frustration.

1.4. Different types of studies on emotional AI in EFL education

Existing studies on emotional AI in EFL education show that research in this field involves different types of studies, such as experimental studies (using a control-group design), field studies (using a one-group design), correlational studies (investigating relationships between different factors or variables), and technological development studies. Different types of studies contribute to knowledge in this field from different perspectives. Identifying the type of an individual study and summarizing the findings from each type of study can help researchers to gain a comprehensive understanding of the field. *Experimental studies* and *field studies* focus on examining the effects of using emotional AI on student learning outcomes. *Correlational studies* focus on the relationships between variables or factors to reveal the factors that may influence the adoption or impact of emotional AI. Technological development studies explored the design, implementation, and testing of AI applications using specific AI techniques. The details are elaborated below.

Experimental studies adopt a control-group design to investigate the effects of AI applications on EFL learning by comparing learning outcomes between experimental and control groups. For example, Nazari et al. (2021) explored the effects of an AI digital writing tool that can provide formative feedback and assessment on student performance. The effects were examined by comparing student learning outcomes between those using this AI tool and other not using the tool.

Field studies employ a one-group design to examine student learning outcomes without a control group for comparison. For example, the study of Ericsson et al. (2023) was conducted with 22 junior secondary students, who were provided with a spoken dialogue system to practice English speaking with personalized feedback; survey questionnaires were used to assess student satisfaction, emotions, and self-estimated ability at the end of the study.

Correlational studies examined the relationships between variables or factors in AI-supported language education, which can reveal the factors that may influence the adoption or impact of emotional AI on EFL education. For example, Du and Gao (2022) collected survey data to figure out the factors that affect teachers' use of AI technology in EFL teaching. Wang et al. (2022) collected survey data and learning data to explore the factors predicting student learning outcomes in AI-supported English-speaking learning.

Technological development studies explored the design and implementation of AI applications using specific AI techniques (e.g., deep learning, machine learning). They were developed to recognize learners' emotional states, for example based on pulse signal analysis (Chen & Lee, 2011) or semantic analysis of students' written comments (Zhou & Gao, 2023). Tian et al. (2023) analyzed students' written comments to detect their affective states in learning EFL. These studies often used public or self-collected datasets to test the technical performance of developed AI applications (Zhang & Wang, 2024). They laid the technological foundation for promoting emotional AI applications in EFL education (Chen et al., 2022).

1.5. Existing reviews

Given a growing number of studies on AI in language education, Huang et al. (2023) conducted a bibliometric analysis of the literature, which revealed popular topics appearing in published studies, such as automated evaluation and error detection, intelligent tutoring systems, personalized systems, conversational agents or chatbots, and affective states and emotions. They also noted that some issues such as data privacy, digital divide, and technology reliability should be addressed in promoting AI in language education. Sharadgah and Sa'di (2022) conducted a systematic review of published studies, reporting promising impacts of AI on English language education. This review also indicated that AI in English language education is still in its infancy especially in some respects such as emotions and body language.

Existing reviews on AI in EFL education have primarily focused on AI-based cognitive support for improving language knowledge and skills (Huang et al., 2023; Sharadgah & Sa'di, 2022). These reviews made few investigations into the emotional affordances of AI applications in language education. While emotional AI in EFL education has attracted increasing attention with a growing number of studies, there is a lack of a systematic literature review that synthesizes existing findings and provides meaningful suggestions for future research and practice in this field.

First, considering the importance of emotion in language learning and an increasing number of published studies on emotional AI in EFL education, it is important to provide researchers and practitioners with a holistic view of research in this emerging field, which can start from the general characteristics of published studies in this field. *Second*, given that individual studies demonstrated different ways of using emotional AI in EFL teaching and learning, researchers and practitioners need to have an integrative view of the affordances of emotional AI applications in order to apply emotional AI in EFL education in effective ways. *Third*, given that individual studies investigated the effects of emotional AI on EFL learning outcomes in different dimensions (behavioral, cognitive, and affective), it is important to summarize the findings to help researchers and practitioners gain a comprehensive understanding of the impacts of emotional AI on EFL education. *Fourth*, considering that individual studies explored different factors (e.g., usefulness of technology,

necessary support for users) that may affect the adoption or effects of emotional AI, there is a need to summarize the findings on such factors in order to make effective use of emotional AI in EFL education.

1.6. The present study

To this end, this study aims to provide a systematic review of published studies on emotional AI, i.e., AI applications that provide emotional support and/or create emotional impacts on student learning, in the field of EFL education. The study aims to answer the research questions presented in Table 1.

To answer RQ1, we summarized the general characteristics of individual studies, including publication year, country or region, study type, education level, learning mode, and learning subject. To address RQ2, we categorized the affordances or functions of emotional AI applications in the EFL education context reported in individual studies. To answer RQ3, we synthesized the effects of emotional AI on student learning outcomes in EFL education reported in experimental and field studies. To address RQ4, we identified the factors that may influence the adoption or effects of emotional AI in EFL education, which are mainly revealed in correlational studies.

2. Method

2.1. Literature search

To identify the literature, we searched the Scopus database for articles published on AI applications that provide emotional support and/or create emotional impacts on student learning in EFL education. The Scopus database was selected since it is the repository with the largest number of articles meeting the quality requirements of scientific peer review (Harzing & Alakangas, 2016; Zhang & Eichmann-Kalwara, 2019). The key terms included in the search query were determined based on published reviews and studies in this field (e.g., Huang et al., 2023; Pokrivcakova, 2019). They included multiple topics including artificial intelligence, English language, education, and emotion, as shown in Table 2. The search query was set up as follows: ("ai" OR "artificial intelligence" OR "intelligent" OR "machine learning" OR "deep learning" OR "neural network" OR "agent*") AND ("mood" OR "emotion*" OR "facial expression" OR "affect*" OR "sentiment") AND ("student" OR "educat*" OR "learner*" OR "tutor*" OR "teach*" OR "instruct*" OR "train*") AND (("language" OR "second language" OR "foreign language" OR "ESL" OR "EFL") AND ("English")). The search was conducted on November 3, 2023, which yielded 931 records.

2.2. Inclusion and exclusion criteria

A set of inclusion and exclusion criteria was set up for selecting articles. Considering that research on English language education is mainly published in English, the language was limited to English. To ensure the quality of selected articles, the document type was restricted to articles published in peer-reviewed academic journals without including conference papers and book chapters. Considering that there was little research on emotional AI in language education before 2001, only articles published since 2001 were selected. Moreover, articles focusing on conceptual discussions or literature reviews without concrete investigations into AI applications in EFL education were excluded.

In short, the articles selected for this systematic review had to meet the following criteria: (1) written in English, (2) published in peer-reviewed journals, (3) published between 2001 and 2023, and (4) exploring AI applications that provide emotional support and/or create emotional or affective impacts for student learning in EFL education.

2.3. Article screening and selection process

The process of article screening and selection was performed based on the PRISMA framework (Moher et al., 2010) as outlined in Fig. 1. *First*, the articles meeting the first three criteria were selected. As a result, 387 articles were obtained for further screening. *Second*, two researchers went through each of these articles to determine whether they met the fourth criterion, with an agreement coefficient of 0.89. After all disagreements were resolved, 366 articles were excluded, i.e., 21 articles remained for analysis. *Third*, other two authors reviewed the selected articles, and four articles were excluded due to quality issues (e.g., no concrete application of AI technologies in teaching or learning practice, lack of adequate details for analysis). *Fourth*, backward and forward searches were

Table 1
Research questions.

Research questions (RQs)		Focus
RQ1:	**What** are the general characteristics of the studies on emotional AI in EFL education?	Publication year, country or region, study type, education level, learning mode, and learning subject
RQ2:	**Why** can emotional AI support EFL education?	Affordances or functions of emotional AI applications used in EFL education
RQ3:	**How** effective is emotional AI in supporting EFL education?	Effects of emotional AI on student learning outcomes in EFL education
RQ4:	**What** factors may affect the impacts of emotional AI on EFL education?	Factors influencing the adoption or effects of emotional AI in EFL education

Table 2
Topics and keywords included in the literature search.

Topic	Keyword
Artificial Intelligence	ai, artificial intelligence, intelligent, machine learning, deep learning, neural network, agent*
Emotion	mood, emotion*, facial expression, affect*, sentiment
Education	student, educat*, learner*, tutor*, teach*, instruct*, train
English language	Language, second language, foreign language, English, ESL, EFL

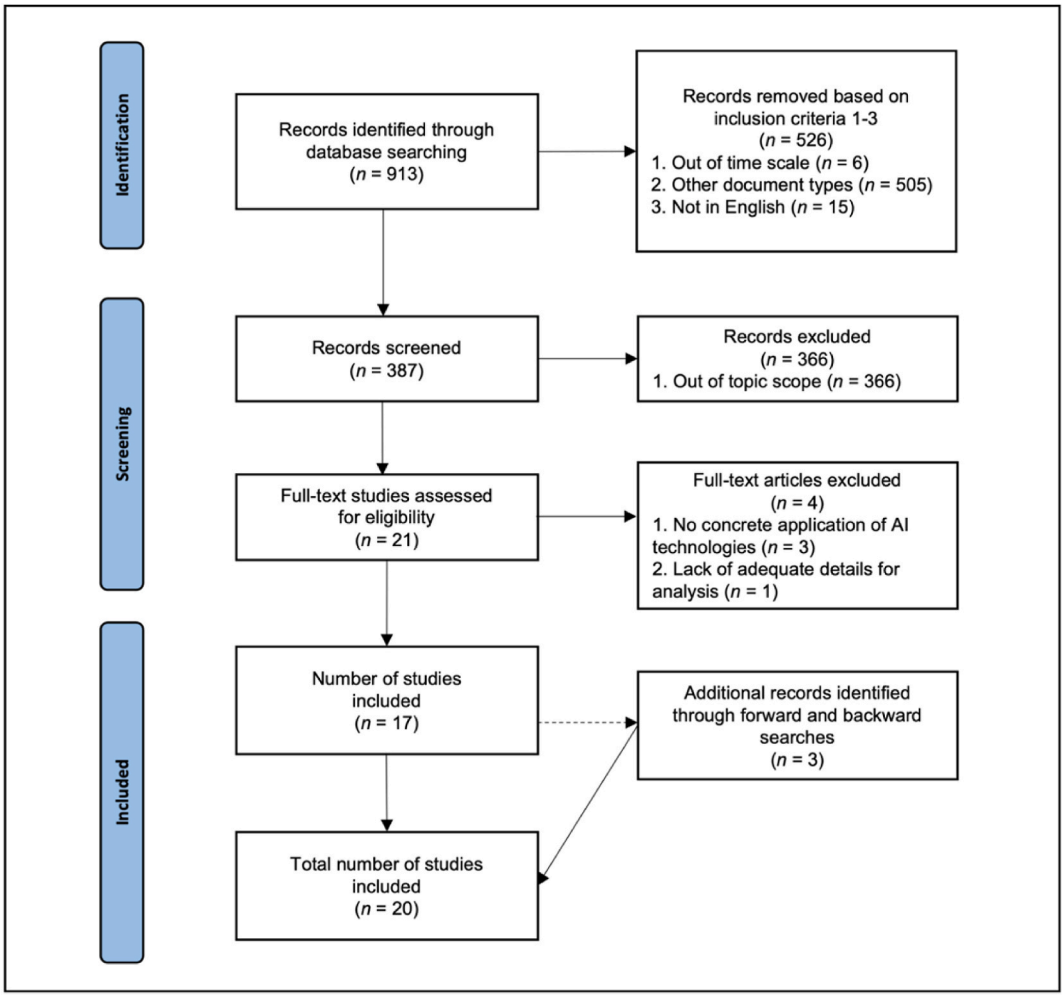

Fig. 1. The PRISMA flow chart for literature search and selection.

performed, through which three more articles meeting the selection criteria were identified. Thus, 20 studies were included in this review.

2.4. Coding framework

The features of each selected study were coded in order to describe and differentiate selected studies and extract findings. The coding framework was established based on prior reviews in the fields of AI in education (Chen et al., 2022), AI in language education (Huang et al., 2023; Sharadgah & Sa'di, 2022), and technology-enhanced learning (Chen et al., 2018; Sikström et al., 2022). For each selected study, we extracted relevant data about its general characteristics, AI affordances, and learning outcomes. The general characteristics of individual studies include several elements, such as when and where the study was conducted, study type, education level, learning modes, and learning subject. Learning outcomes include cognitive, behavioral, and affective outcomes. The details of these elements are elaborated as follows.

Country or region. The codes for country or region refer to where the study was conducted, or the country or region of the

corresponding author or first author if the information about the study location was not reported.

Study type. As mentioned, different types of studies were conducted on emotional AI, including experimental study (using a control-group design), field study (using a one-group design), correlational study, and technological development study, as elaborated in Table 3.

Education level. The codes for education level include elementary, junior secondary, senior secondary, and tertiary (college and university).

Learning mode. The codes for learning mode include classroom learning, online or virtual learning, and mixed learning (Condie & Livingston, 2016). Classroom learning refers to traditional learning in the classroom. Online or virtual learning focuses on learning by interaction with online or virtual learning resources or computer-based systems. Mixed learning includes classroom learning, online or virtual learning, and outside classroom learning.

Learning subject. The learning subjects in EFL education include speaking, listening, vocabulary, grammar, writing, and general English skills (covering listening, speaking, reading, and writing, without a specific focus).

AI affordance. AI applications have been widely applied in language education as reported in previous reviews and studies in this field. These AI applications provide a variety of affordances or offer a variety of facilities to support language teaching and learning, such as the provision of personalized learning content, enabling one-on-one conversations through conversational agents or chatbots, and personalized recommendations or feedback (Huang et al., 2023; Pokrivcakova, 2019; Zhu et al., 2023). Among the 20 articles selected in this study, five affordances in AI applications were identified, which were coded as enabling human-like conversations, providing personalized real-time feedback or instructions, translating images into English text, generating personalized learning content and tasks, and recognizing and analyzing emotions. The details are presented in Section 3.2.

Learning outcomes. Student learning outcomes can be categorized as cognitive, behavioral, and affective outcomes based on the literature (Chen et al., 2018; Wei et al., 2021). Cognitive outcomes focus on academic achievements, i.e., the acquisition of knowledge and skills (e.g., English speaking proficiency, vocabulary knowledge, self-regulation skills). Behavioral outcomes focus on actual behaviors, such as behavioral engagement. Affective learning outcomes focus on attitudes (e.g., acceptance, satisfaction, willingness to communicate in English) and feelings (e.g., enjoyment, self-efficacy, confidence, anxiety), which are often measured by questionnaire surveys.

2.5. Inter-coder reliability

The first three authors discussed the coding framework, which was approved by the other authors of this study. Two trained coders independently performed all the coding. The inter-coder reliability calculated based on Cohen's k was 0.91 for research type, 0.98 for education level, 0.96 for learning subject, 0.95 for country or region, 0.95 for learning mode, and 0.89 for AI applications. All disagreements were resolved through further discussion.

3. Results

3.1. What are the general characteristics of the studies on emotional AI in EFL education?

Publication year. As shown in Fig. 2, most of the selected articles were published in 2022 and 2023.

Country or region. There are more studies conducted in China (N = 6), Taiwan (N = 4) and South Korea (N = 4) than in other countries or regions as show in Fig. 3.

Study type. Among the 20 included studies, four types of studies were identified, including experimental study (6 articles), field study (7 articles), correlational study (2 articles), and technological development study (5 articles). There are six *experimental studies* adopting a control-group design to examine the effects of AI application by comparing student performance between experimental and control groups (Hsu et al., 2023; Kim et al., 2022; Nazari et al., 2021; Nguyen et al., 2020; Sargazi Moghadam et al., 2023; Shadiev et al., 2020). Moreover, there are seven *field studies* adopting a one-group design to explore the implementation of emotional AI in EFL education and evaluate learning outcomes without a control group (Ayedoun et al., 2019; Ericsson et al., 2023; Ericsson & Johansson, 2023; Jeon, 2022; Lee et al., 2011; Rahman & Tomy, 2023; Shin & Lee, 2021).

In addition to examining the effects of AI applications, it is important to explore the factors that may influence the adoption and effects of AI applications, which were investigated in two *correlational studies* (Du & Gao, 2022; Wang et al., 2022). Other five studies focused on *technological development*, i.e., technical design, implementation, and testing of AI applications developed for automatic

Table 3
Types of studies.

Study type	Description
Experimental study (using a control-group design)	Examining the effects of using AI by comparing learning outcomes between experimental and control groups.
Field study (using a one-group design)	Examine the effects of using AI by analyzing students' learning outcomes, often by analyzing posttest data or comparing pretest and posttest data, without a control group.
Correlational study	Examining the relationships between variables or factors in AI-supported language education.
Technological development study	Designing and implementing an AI application using specific AI techniques (e.g., deep learning, machine learning) and testing the performance of the implemented AI application.

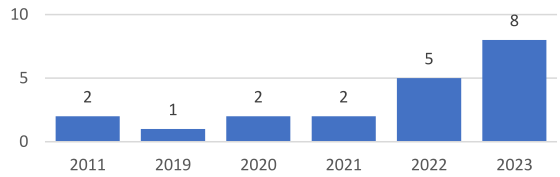

Fig. 2. Number of studies across years.

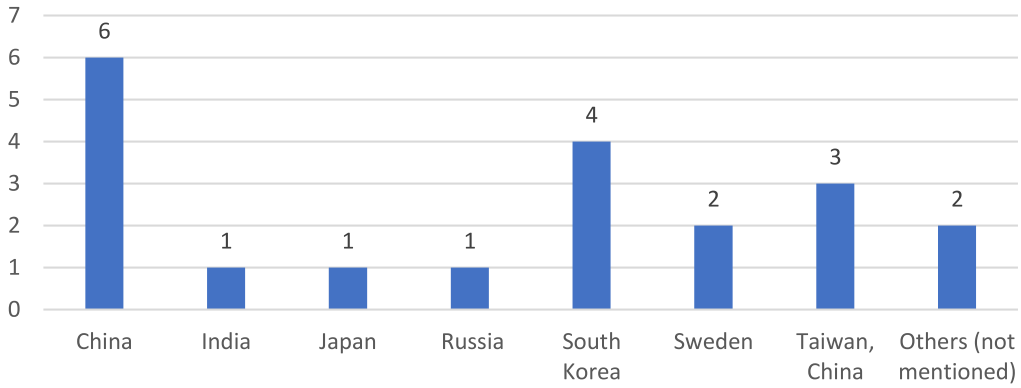

Fig. 3. Number of studies conducted in different countries or regions.

recognition and analysis of emotions in EFL education (Chen & Lee, 2011; Luo, 2022; Tian et al., 2023; Wang, 2023; Zhou & Gao, 2023).

Education level. As shown in Fig. 4, there were five studies conducted with university students, five studies with elementary students, and two studies with junior secondary students. One study was conducted with both senior secondary and university students. In addition, one study did not report on the participants' education level (Sargazi Moghadam et al., 2023). The other six studies not specifying specific education levels were categorized as "N/A".

Learning mode. As shown in Fig. 5, most of the studies adopted online or virtual learning mode (N = 8), and others adopted mixed learning (N = 5) or classroom learning (N = 1) mode. The other six studies not specifying specific learning modes were categorized as "N/A".

Learning subject. As shown in Fig. 6, most of the studies focused on the learning of speaking skills (N = 8) and others focused on the learning of writing, vocabulary, grammar, or general English skills. One study focused on both vocabulary and grammar learning. The other six studies not specifying specific learning subjects were categorized as "N/A".

3.2. Why can emotional AI support EFL education?

To answer this question, we explored the affordances or functions of emotional AI applications in the EFL education context reported in individual studies. As presented in Table 4, five affordances of emotional AI applications were identified from the 20 studies selected in this review. The first three (i.e., enabling human-like conversations, providing personalized real-time feedback or

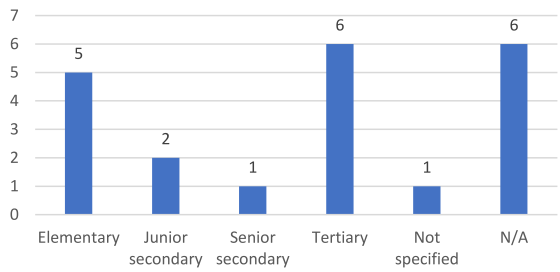

Fig. 4. Number of studies at different education levels.

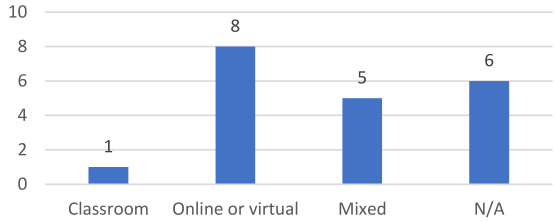

Fig. 5. Number of studies for different learning modes.

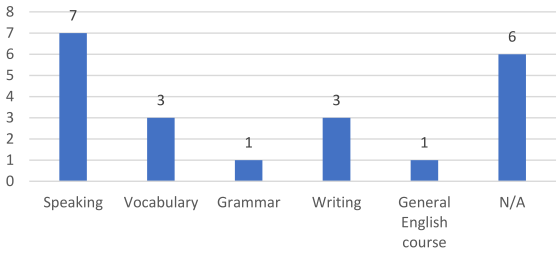

Fig. 6. Number of studies focusing on different learning subjects.

instructions, translating images into English text) focus on supporting student learning in both cognitive and emotional aspects; the fourth one (i.e., generating personalized learning content and tasks) focus on supporting student learning in cognitive aspects; and the purpose of the last one (recognizing and analyzing emotions) is to provide support responding to emotional needs.

Enabling human-like conversations (7 studies). AI applications such as AI chatbots and conversational agents have been increasingly introduced to support language speaking and writing. AI Chatbots or conversational agents or assistants were applied to improve English speaking in six articles (Ayedoun et al., 2019; Ericsson et al., 2023; Ericsson & Johansson, 2023; Jeon, 2022; Lee et al., 2011; Rahman & Tomy, 2023) and to improve English writing in one study (Shin & Lee, 2021). Such kind of AI applications can not only provide learners with personalized conversations in natural language, but also provide personalized interactions and feedback based on the problems or difficulties the learners encounter (Ayedoun et al., 2019; Rahman & Tomy, 2023). Besides, conversational agents can make gestures and facial expressions to reduce student anxiety during the study (Ericsson et al., 2023; Ericsson & Johansson, 2023; Lee et al., 2011).

Providing personalized real-time feedback or instructions (3 studies). To support learners during the learning process, AI technologies have been applied to assess individual performance and analyze individual differences recorded in learner profiles to generate personalized real-time feedback or instructions to learners. Three studies selected in this review adopted such kind of AI applications. Nazari et al. (2021) explored the use of digital writing assistant to provide students with formative feedback to address grammar problems in English academic writing, and reported its effectiveness in improving students' engagement, enjoyment, and self-efficacy and decreasing anxiety. Sargazi Moghadam et al. (2023) integrated system-generated recommendations for micro-break activities into an e-learning system; learners can choose their emotional state in the system; when needed, the system can generate rest activity suggestions for learners to adjust their emotions and reduce anxiety. In the study of Wang et al. (2022), an AI application was

Table 4
Five AI affordances.

AI affordance	Support for student learning		Number of studies
	Cognitive aspect	Emotional aspect	
Enabling human-like conversations	Provide instant responses to queries and coherent conversations in natural language	Generate facial expressions and gestures to regulate emotions	7 (field studies)
Providing personalized real-time feedback or instructions	Provide real-time and individualized feedback or instructions with computer-based assessment	Generate emotion-related recommendations and empathetic instructions	3 (2 experimental studies, 1 correlational study)
Translating images into English text	Recognize images and objects and translate image content into English text	No	3 (experimental studies)
Generating personalized learning content and tasks	Provide personalized learning content and adaptive task	No	2 (1 experimental study, 1 technological development study)
Recognizing and analyzing emotions	No	Identify emotions and provide relevant support	5 (technological development studies)

incorporated to improve English speaking by using AI to read the sentence, record students' reading, identify pronunciation problems, and give instructions to correct the problems.

Translating images into English text (3 studies). Using pictures to teach vocabulary is a commonly used strategy in English education. In recent years, AI-powered image-to-text recognition technology has been developed and used to automatically recognize images and objects and generate English text (such as vocabularies, phrases, and sentences) to describe the content of recognized images. In this review, three studies explored the use of this technology to improve vocabulary acquisition and writing skills. In the study of Shadiev et al. (2020) and Hsu et al. (2023), students learned vocabulary by taking pictures of objects and getting the names of those objects in English. Nguyen et al. (2020) applied this technology to improve students' writing skills by asking them to take pictures in a real-life environment, get AI-generated vocabulary describing the pictures, and write an essay about the environment.

Generating personalized learning content and task (2 studies). Personalized and adaptive learning systems such as intelligent tutoring systems have been used to afford personalized learning experiences (Chen et al., 2022). In this review, two studies applied AI-based tools to support personalized learning. Kim et al. (2022) applied an intelligent tutoring system to support student learning of grammar and vocabulary by providing individualized learning content and adaptive tasks for self-regulated learning. Wang (2023) developed an AI application to recognize learners' speech emotions and recommend culture-related content for learners when they have difficulties in communication due to cultural differences.

Recognizing and analyzing emotions (5 studies). To provide emotional support to learners during the study, it is important to recognize or analyze their emotional states. Five studies (Chen & Lee, 2011; Luo, 2022; Wang, 2023; Zhou & Gao, 2023) included in this review focused on the technological development of AI applications capable of recognizing learners' emotional states by analyzing emotion-related data (e.g., pulse signal analysis, semantic analysis of students' written comments).

3.3. How effective is emotional AI in supporting EFL education?

The effects of AI applications on EFL education varied with the affordances of AI applications. Based on the findings reported in the selected studies, Table 5 summarizes the effects of different AI affordances in terms of student learning outcomes in behavioral, cognitive, and behavioral aspects. Most studies, including those selected in this review, have focused on analyzing cognitive and affective outcomes rather than behavioral outcomes.

Effects of enabling human-like conversations. As mentioned, there were six studies exploring the use of AI-based conversational agents to support English speaking and writing. They reported positive effects of this AI application on *affective learning outcomes* in terms of students' acceptance and satisfaction with AI-based conversational agent, their willingness to communicate in English, their enjoyment and reduced anxiety when learning with AI-based conversational agent, activated interest in learning, and self-efficacy or confidence in speaking and writing skills (Ericsson et al., 2023; Ericsson & Johansson, 2023; Jeon, 2022; Lee et al., 2011; Rahman & Tomy, 2023; Shin & Lee, 2021). The findings on cognitive outcomes reveal the positive effects of using AI-based conversational agents on students' speaking proficiency and vocabulary knowledge (Lee et al., 2011; Shin & Lee, 2021).

Effects of providing personalized real-time feedback or instructions. As mentioned, three studies investigated the use of system-generated feedback or instructions to support English writing, speaking and general learning in an online environment. The studies of Nazari et al. (2021) and Sargazi Moghadam et al. (2023) found positive effects of such kind of AI applications on affective learning outcomes involving satisfaction, engagement, self-efficacy, and enjoyment. Moreover, Sargazi Moghadam et al. (2023) reported positive effects in improving students' general English skills (listening, speaking, reading, and reading). The study of Wang et al. (2022) explored the factors predicting student learning achievements in the context of using AI to support English speaking by identifying and correcting students' pronunciation problems. They found that language enjoyment and teaching presence positively and negatively predict learning outcomes in this context, respectively.

Effects of translating images into English text. The studies of Hsu et al. (2023) and Shadiev et al. (2020) reported positive effects of using the image-to-text recognition technology on improving students' vocabulary knowledge. Shadiev et al. (2020) also demonstrated the effects of using this tool on improving student satisfaction, which may directly influence their emotional experience. Another study by Nguyen et al. (2020) reported positive learning outcomes from using this AI application, that is, improving students' writing skills and confidence and reducing anxiety in using English.

Table 5
AI affordances and their effects on student learning.

AI affordance	Effects on student learning outcomes		
	Behavioral	Cognitive	Affective
Enabling human-like conversations	Not reported	Vocabulary knowledge, speaking proficiency	Satisfaction, Acceptance, Willingness to communicate in English, Interest in English learning, Enjoyment, Reduced anxiety, Self-efficacy
Providing personalized real-time feedback or instructions	Engagement	General English skills (listening, speaking, and reading)	Self-efficacy, Enjoyment, Satisfaction
Translating images into English text	Engagement	Vocabulary knowledge, Writing skills	Satisfaction, Lower anxiety, Confidence
Generating personalized learning content and tasks	Not reported	Not reported	Satisfaction
Recognizing and analyzing emotions	Not reported	Not reported	Not reported

Effects of generating personalized learning content and task. The study of Wang (2023) developed an AI application that can recommend relevant content to learners after recognizing their speech emotions; the effects of this AI application on student learning outcomes were not investigated. The study of Kim et al. (2022) reported that when using an AI-based system that provided learners with personalized learning experiences based on individual differences, students found that the teacher's cognitive support rather than the teacher's emotional support could effectively support their learning in such a context.

Effects of recognizing and analyzing emotions. There are five studies exploring the technological development of AI applications that can recognize or analyze student emotions. These studies focused on technological development, not examining the effects of such applications on student learning outcomes.

3.4. What factors may affect the impacts of emotional AI on EFL education?

Two studies examined the relationships between variables or factors in the context of AI-supported language education. They revealed the factors that may influence the adoption and effects of emotional AI on EFL education. Du and Gao (2022) found that the effectiveness and efficiency of AI technology are the determining factors that affect teachers' adoption of AI technology. Wang et al. (2022) explored the factors predicting student learning achievements in the context of using AI to support English speaking by identifying and correcting students' pronunciation problems. They found that enjoyment and teaching presence positively and negatively predict learning outcomes in this context, respectively.

4. Discussion

4.1. What are the general characteristics of the studies on emotional AI in EFL education?

The findings of this review show that emotional AI applications in EFL education have attracted increased attention in recent years especially since 2022. Among the articles selected in this review, most studies were conducted in China and other countries or regions in Asia. The result is consistent with the wave of AI education in Asia and the government-initiated informatization policies in China (Yan & Yang, 2021) in addition to the demand for improving EFL education in Asia countries.

Regarding *study type*, this review included six articles on experimental studies and seven articles on field studies, which represent the main research theme in this field, that is, examining the effects of emotional AI on EFL students' learning outcomes. Besides, five articles on technological development of emotional AI applications for EFL education and two articles on correlational studies in AI-supported EFL education were included in the review.

Regarding *education level*, most studies were conducted with elementary, secondary (junior and senior), and college students, echoing the widespread promotion of technology-enabled language education at elementary, secondary, and tertiary levels (Huang et al., 2023).

Regarding *learning mode*, more studies were conducted in online or virtual and mixed learning environments than in the classroom, which is consistent with the review by Sharadgah and Sa'di (2022) that AI applications for EFL education are often incorporated in technology-rich environments.

Regarding *learning subject*, more studies were conducted to support the learning of English speaking, writing, and vocabulary. This echoes the growing interest in using AI-based conversational agents or chatbots to support English speaking learning and in using AI-enabled translation of images into English text to support the learning of vocabulary and writing.

4.2. Why can emotional AI support EFL education?

As shown in Section 3.2, five AI affordances are identified from the 20 studies selected for this review. Among them, *enabling human-like conversations* was most frequently used to support language learning, mainly for English speaking and writing. The result echoes the important role of AI-based conversational agents or chatbots in English language learning emphasized in previous reviews (Huang et al., 2023; Pokrivcakova, 2019; Yang & Kyun, 2022). The frequent use of conversational agents can be attributed to their salient advantage in reducing speech-related anxieties and enhancing students' willingness to communicate by providing personalized conversations in natural language and instant feedback to learners when they experience problems or difficulties (Ayedoun et al., 2019; Ericsson et al., 2023; Ericsson & Johansson, 2023; Rahman & Tomy, 2023; Shin & Lee, 2021).

Other two AI affordances, *providing personalized real-time feedback or instructions* and *translating images into English text,* have been increasingly applied in EFL education, for example providing students with formative feedback to address grammar problems (Nazari et al. (2021) and helping students learn vocabulary by taking pictures of objects and getting AI-generated English words describing those pictures (Hsu et al., 2023). The findings echo the importance of providing personalized adaptive support to learners during the learning process (Chen et al., 2022) and utilizing pictures to support vocabulary acquisition in EFL education (Elliott & Zhang, 1998).

As reported in the empirical studies included in this review, most emotional AI applications focused on *cognitive* support for language learning by utilizing the five AI affordances mentioned above; some applications integrate cognitive and emotional support, such as using conversational agents to make gestures and facial expressions (Ericsson et al., 2023; Ericsson & Johansson, 2023; Lee et al., 2011) and using emotion-related recommendations and empathetic instructions to regulate emotions (Chen & Lee, 2011; Sargazi Moghadam et al., 2023) in addition to supporting the cognitive process. It was found that none of these applications focused solely on emotional support while ignoring cognitive support; instead, emotional support is often integrated with cognitive support in these studies. The identified pattern echoes the findings of prior research, which indicated that while AI-based emotional support is

conductive to improving learners' motivation and emotions, providing emotional support alone is not effective enough to improve learners' performance (Arroyo et al., 2011; Chen et al., 2012; Kim et al., 2007).

4.3. How effective is emotional AI in supporting EFL education?

The experimental studies included in this review reported promising effects of AI applications on improving students' cognitive and affective learning outcomes in EFL education. The effects varied with different AI affordances. In particular, three AI affordances (enabling human-like conversations, providing personalized real-time feedback or instructions, and translating images into English text) yielded significant effects on students' academic achievements and affective learning outcomes. The promising effects of the three AI affordances can be attributed to their advantages in supporting language learning. To be specific, AI-enabled human-like conversations can reduce speech-related anxieties and enhance students' willingness to communicate to improve speaking and writing skills (Ayedoun et al., 2019; Yang & Kyun, 2022). Providing personalized real-time feedback or instructions can help students solve problems in a timely manner, thereby reducing their anxiety and increasing their engagement, enjoyment, and self-efficacy (Nazari et al., 2021). Using AI-enabled translation of images into English text can support the learning of vocabulary and writing, improving confidence, and reducing anxiety in using English (Hsu et al., 2023; Nguyen et al., 2020; Shadiev et al., 2020).

It is noted that AI applications involving cognitive support alone can enhance both affective and cognitive learning outcomes (Ericsson et al., 2023; Hsu et al., 2023; Nazari et al., 2021; Rahman & Tomy, 2023; Sargazi Moghadam et al., 2023; Shin & Lee, 2021). The finding can be explained by the reciprocal relationships between affective and cognitive processes (Pekrun et al., 2017). When working on a complex task, students often experience difficulties; further, they may experience negative emotions like anxiety or frustration, which can affect their learning achievements. Research shows that the approaches that support students' cognitive processes in complex tasks can improve students' task performance as well as reduce their anxiety and frustration (Corbalan et al., 2009; Wang et al., 2018). The findings are consistent with the self-efficacy theory, in which self-efficacy is generally defined as one's belief in one's ability to successfully execute or complete a specific task (Bandura, 1997). With AI-based support to improve their cognitive processes, students may feel that their abilities have been improved, which can boost their self-efficacy and other positive emotions. In another study included in this review, it was found that when using AI to provide personalized learning experiences, students felt that the teacher's cognitive support rather than the teacher's emotional support can effectively support their learning in such contexts (Kim et al., 2022).

It is also noted that although conversational agents or chatbots are more frequently applied in EFL education, their effects on student learning were only explored in field studies (using a one-group design), without using experimental studies (using a control-group design) to examine the effects more rigorously by comparing learning outcomes between students using and not using conversational agents or chatbots.

4.4. What factors may affect the impacts of emotional AI on EFL education?

In addition to examining the impacts of AI applications, it is important to explore the factors that may influence the adoption and effects of AI applications, as revealed by two correlational studies included in this review. The study of Du and Gao (2022) revealed that attention should be paid to the effectiveness and efficiency of AI technology, which may affect teachers' adoption of AI technology. The study by Wang et al. (2022) found that language enjoyment may positively affect learning outcomes, while teaching presence may negatively affect learning outcomes when using AI to address students' pronunciation problems. The findings implied the importance of fostering enjoyable and student-centered learning environments when incorporating AI applications in EFL education.

5. Limitations

There are several limitations in this study. First, this review collected relevant articles from the Scopus database. While the Scopus database is recognized as the repository with the largest number of articles meeting the quality requirements of scientific peer review (Harzing & Alakangas, 2016; Zhang & Eichmann-Kalwara, 2019), some relevant studies might not be included in this database and thus not included in this review. Second, this study focused on emotional AI in EFL education, an emerging field attracting increasing attention but currently with a limited number of published studies. The small number of articles that met the selection criteria for this review may limit the findings to some extent. Third, due to the small number of included articles, this review has focused on qualitative analysis of the selected studies. Further quantitative analysis of the findings may be conducted when there is a sufficient number of publications in this field.

Meanwhile, there are some issues in existing studies of this field. First, the findings from some studies are limited by the small sample size and specific context of these studies. Second, conclusions drawn from merely self-reported data in some studies are limited because of insufficient evidence on academic achievements. Third, the effectiveness of emotional AI applications in EFL education is subject to other factors beyond technology. More research is needed to examine the factors that may influence the adoption and effects of AI applications in education. This is crucial to effective promotion of AI applications in English language education.

6. Conclusion

This paper presents a systematic review of 20 published studies on emotional AI in the context of EFL education. The main

contribution or findings of this study are summarized as follows. First, this review reveals five affordances of emotional AI applications in EFL education, namely (a) enabling human-like conversations, (b) providing personalized real-time feedback or instructions, (c) translating images into English text, (d) generating personalized learning content and tasks, and (e) recognizing and analyzing emotions. Second, the results show that the first three AI affordances are more frequently offered in AI applications for EFL education. They provide cognitive and emotional support for EFL learners and have shown promising effects on improving students' behavioral, cognitive and affective learning outcomes.

Third, the findings indicate that emotional support is often integrated with cognitive support in AI applications; providing emotional support alone may not be enough to support student learning. On the other hand, providing cognitive support alone can enhance both affective and cognitive learning outcomes. These findings echo the close relationship between cognition and emotion in AI-supported EFL education.

Fourth, the effects of using conversational agents or chatbots were mainly explored in field studies; more experimental studies are expected to examine the effects more rigorously by comparing learning outcomes between students using and not using such applications. Finally, more attention should be paid to the factors that may influence the adoption and effects of emotional AI in EFL teaching and learning. For example, the effectiveness and efficiency of AI technology may affect teachers' adoption of AI in EFL education. Also, it is important to promote language enjoyment and student-centered learning when incorporating AI in EFL education.

Hopefully, the findings of this study may contribute to the existing body of knowledge on AI-assisted EFL education. The findings may provide meaningful insights for researchers, educators, and policymakers to explore the potential of AI in EFL education from both cognitive and emotional perspectives.

CRediT authorship contribution statement

Yuhan Liu: Writing – original draft, Formal analysis, Data curation. **Heng Zhang:** Writing – original draft, Formal analysis, Data curation. **Meilin Jiang:** Writing – original draft, Formal analysis, Data curation. **Juanjuan Chen:** Writing – review & editing, Writing – original draft, Validation. **Minhong Wang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Acknowledgements

This study was supported by the National Natural Science Foundation of China (Projects No. 61977023 & No. 62377042) and the National Social Science Foundation of China (Project No. BCA230274). The authors would like to thank the anonymous reviewers and the editor for their helpful comments and suggestions. The authors would also thank Professor Haijing Jiang for his valuable guidance support for this study.

References

Arroyo, I., Woolf, B. P., Cooper, D. G., Burleson, W., & Muldner, K. (2011). The impact of animated pedagogical agents on girls' and boys' emotions, attitudes, behaviors and learning. In *Proceedings of IEEE 11th international conference on advanced learning technologies* (pp. 506–510). IEEE. https://doi.org/10.1109/ICALT.2011.157.

Ayedoun, E., Hayashi, Y., & Seta, K. (2019). Adding communicative and affective strategies to an embodied conversational agent to enhance second language learners' willingness to communicate. *International Journal of Artificial Intelligence in Education, 29*, 29–57. https://doi.org/10.1007/s40593-018-0171-6

Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman and Company.

Brand, S., Reimer, T., & Opwis, K. (2007). How do we learn in a negative mood? Effects of a negative mood on transfer and learning. *Learning and Instruction, 17*(1), 1–16. https://doi.org/10.1016/j.learninstruc.2006.11.002

Chen, L., Chen, P., & Lin, Z. (2020). Artificial intelligence in education: A review. *IEEE Access, 8*, 75264–75278. https://doi.org/10.1109/ACCESS.2020.2988510

Chen, C.-M., & Lee, T.-H. (2011). Emotion recognition and communication for reducing second-language speaking anxiety in a web-based one-to-one synchronous learning environment. *British Journal of Educational Technology, 42*(3), 417–440. https://doi.org/10.1111/j.1467-8535.2009.01035.x

Chen, G. D., Lee, J. H., Wang, C. Y., Chao, P. Y., Li, L. Y., & Lee, T. Y. (2012). An empathic avatar in a computer-aided learning program to encourage and persuade learners. *Educational Technology & Society, 15*(2), 62–72. http://www.jstor.org/stable/jeductechsoci.15.2.62.

Chen, J., Wang, M., Kirschner, P. A., & Tsai, C. C. (2018). The role of collaboration, computer use, learning environments, and supporting strategies in CSCL: A meta-analysis. *Review of Educational Research, 88*(6), 799–843. https://doi.org/10.3102/0034654318791584

Chen, X., Zou, D., Xie, H., Cheng, G., & Liu, C. (2022). Two decades of artificial intelligence in education: Contributors, collaborations, research topics, challenges, and future directions. *Educational Technology & Society, 25*(1), 28–47.

Condie, R., & Livingston, K. (2016). Blending online learning with traditional approaches: Changing practices. *British Journal of Educational Technology, 38*(2), 337–348. https://doi.org/10.1111/j.1467-8535.2006.00630.x

Corbalan, G., Kester, L., & Van Merriënboer, J. J. G. (2009). Dynamic task selection: Effects of feedback and learner control on efficiency and motivation. *Learning and Instruction, 19*(6), 455–465. https://doi.org/10.1016/j.learninstruc.2008.07.002

Du, Y., & Gao, H. (2022). Determinants affecting teachers' adoption of AI-based applications in EFL context: An analysis of analytic hierarchy process. *Education and Information Technologies, 27*(7), 9357–9384. https://doi.org/10.1007/s10639-022-11001-y

Elliott, R. T., & Zhang, Q. (1998). Interference in learning context-dependent words. *Educational Psychology, 18*(1), 5–25.

Ericsson, E., & Johansson, S. (2023). English speaking practice with conversational AI: Lower secondary students' educational experiences over time. *Computers and Education: Artificial Intelligence, 5*, Article 100164. https://doi.org/10.1016/j.caeai.2023.100164

Ericsson, E., Lundin, J., & Sofkova Hashemi, S. (2023). From deadpan machine to relating socially: Middle school students' experiences speaking English with embodied conversational agents. *Journal of Research on Technology in Education*. https://doi.org/10.1080/15391523.2023.2213902

Feidakis, M. (2016). A review of emotion-aware systems for e-learning in virtual environments. In S. Caballé, & R. Clarisó (Eds.), *Formative assessment, learning data analytics and gamification: In ICT education* (pp. 217–242). Elsevier.

Harzing, A. W., & Alakangas, S. (2016). Google scholar, Scopus and the web of science: A longitudinal and cross-disciplinary comparison. *Scientometrics, 106*, 787–804. https://doi.org/10.1007/s11192-015-1798-9

Hsiao, J.-C., & Chang, J. S. (2023). Enhancing EFL reading and writing through AI-powered tools: Design, implementation, and evaluation of an online course. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2023.2207187

Hsu, T.-C., Chang, C., & Jen, T.-H. (2023). Artificial intelligence image recognition using self-regulation learning strategies: Effects on vocabulary acquisition, learning anxiety, and learning behaviours of English language learners. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2023.2165508

Hu, D., Wang, M., Huang, L., & De La Torre, J. (2024). Supporting content and language integrated learning through computer-based dual concept mapping. Computer Assisted Language Learning, 1-28. doi:10.1080/09588221.2024.2317842.

Huang, W., Hew, K. F., & Fryer, L. K. (2022). Chatbots for language learning—are they really useful? A systematic review of chatbot-supported language learning. *Journal of Computer Assisted Learning, 38*(1), 237–257. https://doi.org/10.1111/jcal.12610

Huang, X., Zou, D., Cheng, G., Chen, X., & Xie, H. (2023). Trends, research issues and applications of artificial intelligence in language education. *Educational Technology & Society, 26*(1), 112–131. https://doi.org/10.30191/ETS.202301_26(1).0009

Jeon, J. (2022). Exploring AI chatbot affordances in the EFL classroom: Young learners' experiences and perspectives. *Computer Assisted Language Learning, 37*(1–2), 1–26. https://doi.org/10.1080/09588221.2021.2021241

Kim, Y., Baylor, A. L., & Shen, E. (2007). Pedagogical agents as learning companions: The impact of agent emotion and gender. *Journal of Computer Assisted Learning, 23*(3), 220–234. https://doi.org/10.1111/j.1365-2729.2006.00210.x

Kim, H.-S., Kim, N.-Y., & Cha, Y. (2022). A study on the use of AI-based learning programs by EFL students with different types of teacher support. *Korean Journal of English Language and Linguistics, 22*, 355–376. https://doi.org/10.15738/kjell.22..202204.355

Lee, S., Noh, H., Lee, J., Lee, K., Lee, G. G., Sagong, S., & Kim, M. (2011). On the effectiveness of robot-assisted language learning. *ReCALL, 23*(1), 25–58. https://doi.org/10.1017/S0958344010000273

Luo, Y. (2022). Research on multifeature intelligent correction of spoken English. *Computational Intelligence and Neuroscience, 2022*, Article 8241241. https://doi.org/10.1155/2022/8241241

Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & PRISMA Group. (2010). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *International Journal of Surgery, 8*(5), 336–341. https://doi.org/10.1016/j.ijsu.2010.02.007

Nazari, N., Shabbir, M. S., & Setiawan, R. (2021). Application of artificial intelligence powered digital writing assistant in higher education: Randomized controlled trial. *Heliyon, 7*, Article 07014. https://doi.org/10.1016/j.heliyon.2021.e07014

Nguyen, T. H., Hwang, W. Y., Pham, X. L., & Pham, T. (2020). Self-experienced storytelling in an authentic context to facilitate EFL writing. *Computer Assisted Language Learning, 35*(4), 666–695. https://doi.org/10.1080/09588221.2020.1744665

Oxford, R. (2015). Emotion as the amplifier and the primary motive: Some theories of emotion with relevance to language learning. *Studies in Second Language Learning and Teaching, 5*(3), 371–393.

Pekrun, R., Lichtenfeld, S., Marsh, H. W., Murayama, K., & Goetz, T. (2017). Achievement emotions and academic performance: Longitudinal models of reciprocal effects. *Child Development, 88*(5), 1653–1670. https://doi.org/10.1111/cdev.12704

Phelps, E. A. (2006). Emotion and cognition: Insights from studies of the human amygdala. *Annual Review of Psychology, 57*, 27–53. https://doi.org/10.1146/annurev.psych.56.091103.070234

Plonsky, L., Sudina, E., & Teimouri, Y. (2022). Language learning and emotion. *Language Teaching, 55*(3), 346–362. https://doi.org/10.1017/S0261444821000434

Pokrivcakova, S. (2019). Preparing teachers for the application of AI-powered technologies in foreign language education. *Journal of Language and Cultural Education, 7*(3), 135–153. https://doi.org/10.2478/jolace-2019-0025

Rahman, A., & Tomy, P. (2023). Intelligent personal assistant-an interlocutor to mollify foreign language speaking anxiety. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2023.2204324

Rus, V., D'Mello, S., Hu, X., & Graesser, A. (2013). Recent advances in conversational intelligent tutoring systems. *AI Magazine, 34*(3), 42–54. https://doi.org/10.1609/aimag.v34i3.2485

Russell, S. J., & Norvig, P. (2016). *Artificial intelligence: A modern approach*. Pearson Education Limited.

Sargazi Moghadam, T., Darejeh, A., Delaramifar, M., & Mashayekh, S. (2023). Toward an artificial intelligence-based decision framework for developing adaptive e-learning systems to impact learners' emotions. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2023.2188398

Shadiev, R., Wu, T. T., & Huang, Y. M. (2020). Using image-to-text recognition technology to facilitate vocabulary acquisition in authentic contexts. *ReCALL, 32*(2), 195–212. https://doi.org/10.1017/S0958344020000038

Sharadgah, T. A., & Sa'di, R. A. (2022). A systematic review of research on the use of artificial intelligence in English language teaching and learning (2015-2021): What are the current effects? *Journal of Information Technology Education: Research, 21*, 337–377. https://doi.org/10.28945/4999

Shin, D., & Lee, J. H. (2021). Exploring the use of an artificial intelligence chatbot as second language conversation partners. 영어학, *21*, 375–391. https://doi.org/10.15738/kjell.21..202104.375

Sikström, P., Valentini, C., Sivunen, A., & Kärkkäinen, T. (2022). How pedagogical agents communicate with students: A two-phase systematic review. *Computers & Education, 188*, Article 104564. https://doi.org/10.1016/j.compedu.2022.104564

Smutny, P., & Schreiberova, P. (2020). Chatbots for learning: A review of educational chatbots for the Facebook messenger. *Computers & Education, 151*, Article 103862. https://doi.org/10.1016/j.compedu.2020.103862

Strobl, C., Ailhaud, E., Benetos, K., Devitt, A., Kruse, O., Proske, A., & Rapp, C. (2019). Digital support for academic writing: A review of technologies and pedagogies. *Computers & Education, 131*, 33–48. https://doi.org/10.1016/j.compedu.2018.12.005

Tian, X., Alassafi, M. O., & Alsaadi, F. E. (2023). An efficient English teaching driven by enterprise-social media big data: A neural network-based solution. *FRACTALS (fractals), 31*(6), Article 2340151. https://doi.org/10.1142/S0218348X23401515

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist, 46*(4), 197–221.

Wang, F. (2023). AI-based English teaching cross-cultural fusion mechanism. *Evolutionary Intelligence, 16*(5), 1461–1467. https://doi.org/10.1007/s12065-022-00733-y

Wang, X., Pang, H., Wallace, M. P., Wang, Q., & Chen, W. (2022). Learners' perceived ai presences in AI-supported language learning: A study of AI as a humanized agent from community of inquiry. *Computer Assisted Language Learning*. https://doi.org/10.1080/09588221.2022.2056203

Wang, M., Yuan, B., Kirschner, P. A., Kushniruk, A. W., & Peng, J. (2018). Reflective learning with complex problems in a visualization-based learning environment with expert support. *Computers in Human Behavior, 87*, 406–415. https://doi.org/10.1016/j.chb.2018.01.025

Wei, X., Saab, N., & Admiraal, W. (2021). Assessment of cognitive, behavioral, and affective learning outcomes in massive open online courses: A systematic literature review. *Computers & Education, 163*, Article 104097. https://doi.org/10.1016/j.compedu.2020.104097

Xu, J. (2018). Emotion regulation in mathematics homework: An empirical study. *The Journal of Educational Research, 111*(1), 1–11. https://doi.org/10.1080/00220671.2016.1175409

Yan, S., & Yang, Y. (2021). Education informatization 2.0 in China: Motivation, framework, and vision. *ECNU Review of Education, 4*(2), 410–428. https://doi.org/10.1177/2096531120944929

Yang, H., & Kyun, S. (2022). The current research trend of artificial intelligence in language learning: A systematic empirical literature review from an activity theory perspective. *Australasian Journal of Educational Technology, 38*(5), 180–210. https://doi.org/10.14742/ajet.7492

Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education–where are the educators? *International Journal of Educational Technology in Higher Education, 16*, 39. https://doi.org/10.1186/s41239-019-0171-0

Zhang, H., & Wang, M. (2024). A novel deep learning model to improve the recognition of students' facial expressions in online learning environments. *Knowledge Management & E-Learning, 16*(1), 134–150. https://doi.org/10.34105/j.kmel.2024.16.006

Zhang, L., & Eichmann-Kalwara, N. (2019). Mapping the scholarly literature found in Scopus on "research data management": A bibliometric and data visualization approach. *Journal of Librarianship and Scholarly Communication, 7*(1). https://doi.org/10.7710/2162-3309.2266

Zhou, W., & Gao, B. (2023). Construction and application of English-Chinese multimodal emotional corpus based on artificial intelligence. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2023.2169526

Zhu, C., Sun, M., Luo, J., Li, T., & Wang, M. (2023). How to harness the potential of ChatGPT in education? *Knowledge Management & E-Learning, 15*(2), 133–152. https://doi.org/10.34105/j.kmel.2023.15.008